

## ASG Airlines – Dataset & Pipeline Documentation

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### 1. Project Overview

This document describes the completed ASG Airlines end-to-end data engineering pipeline. The implementation uses Databricks Free Edition, Databricks Volumes and Power BI. The supplied Excel workbook is ingested, profiled, cleaned, transformed, privacy-protected and converted into Gold analytical datasets for reporting.

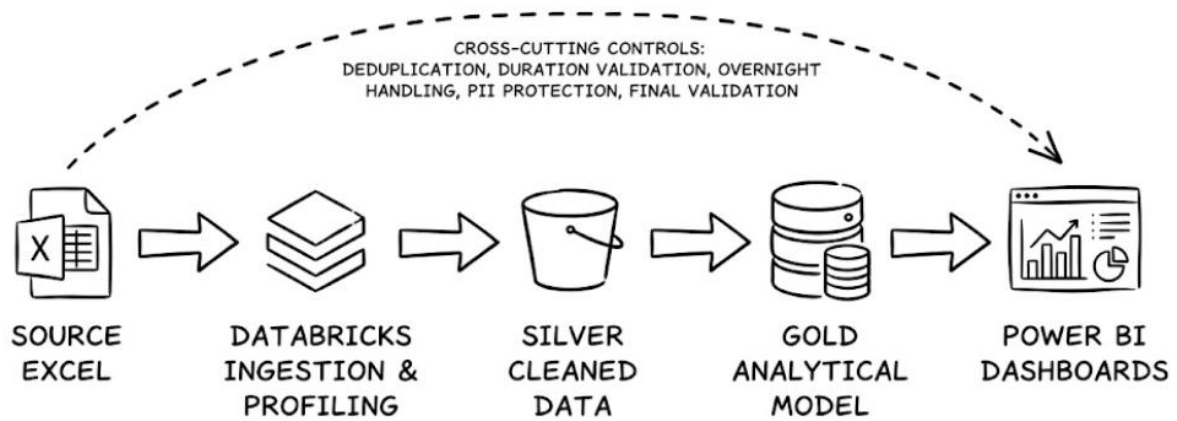
### 2. Dataset Structure

The source workbook contains four primary datasets:

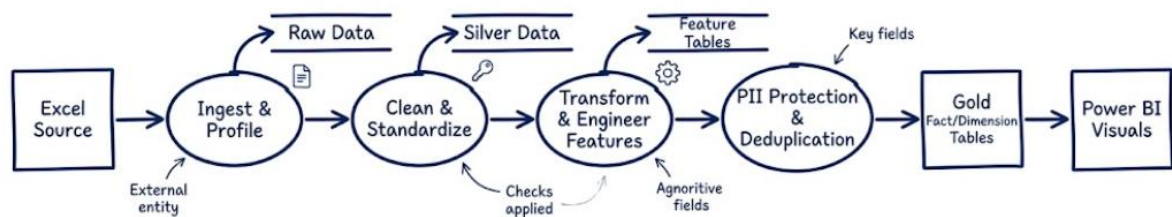
| Dataset    | Raw rows | Purpose                    | Key role                                                             |
|------------|----------|----------------------------|----------------------------------------------------------------------|
| Flights    | 1,020    | Operational flight records | flight_id, airline, source, destination, departure/arrival, duration |
| Bookings   | 1,000    | Booking transactions       | booking_id, flight_id, passenger_id, booking status                  |
| Passengers | 1,039    | Passenger reference data   | passenger identifiers and passenger attributes                       |
| Payments   | 1,000    | Payment transactions       | payment_id, booking_id, amount, payment method, validity             |

### 3. Architecture Diagram

Raw source - Databricks ingestion/profileing - Silver cleaned data - Gold analytical model - Power BI.



#### 4. Data Flow Diagram (DFD)



#### 5. Data Model

Gold outputs are fact\_flights (1,004), fact\_bookings (1,000), fact\_payments (1,000), dim\_route (30), dim\_airline (6), dim\_date (369) and dim\_passenger (1,039).

| Table         | Key / relationship               | Purpose                                                     |
|---------------|----------------------------------|-------------------------------------------------------------|
| fact_flights  | flight_id (PK)                   | Flight-level measures including duration and overnight flag |
| fact_bookings | booking_id (PK), flight_id (FK)  | Booking status and booking-to-flight analysis               |
| fact_payments | payment_id (PK), booking_id (FK) | Payment amount, method and validity                         |
| dim_route     | route_key (PK)                   | Source/destination route analysis                           |
| dim_airline   | airline_key (PK)                 | Airline analysis                                            |
| dim_date      | date_key (PK)                    | Calendar/date analysis                                      |

|               |                    |                                    |
|---------------|--------------------|------------------------------------|
| dim_passenger | passenger_key (PK) | Protected passenger reference data |
|---------------|--------------------|------------------------------------|

Power BI relationships implemented: fact\_flights[flight\_id] 1→\* fact\_bookings[flight\_id]; fact\_bookings[booking\_id] 1→\* fact\_payments[booking\_id].

## 6. Preprocessing and Transformation Logic

| Step                               | Logic                                                                                                                                                           |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Ingestion                       | Load the supplied Excel workbook into Databricks and inspect schemas and record counts.                                                                         |
| 2. Profiling                       | Check nulls, duplicates, data types, keys, timestamps, categorical values and payment fields.                                                                   |
| 3. Flight deduplication            | Remove 15 exact duplicate rows and handle 1 additional duplicate flight_id; 16 flight rows removed overall.                                                     |
| 4. Standardization                 | Normalize data types, text/category values, timestamps, dates and numeric fields.                                                                               |
| 5. Duration calculation/validation | Derive or validate duration in minutes from departure and arrival timestamps while respecting calendar-day boundaries.                                          |
| 6. Overnight handling              | Set is_overnight when arrival occurs on a later calendar date. Final result: 122 overnight flights.                                                             |
| 7. Duration correction             | Apply the implemented duration correction rule where required. Final result: 1 correction and 0 extreme-duration anomalies.                                     |
| 8. Booking preparation             | Deduplicate passenger mapping by passenger_id before joining, preventing many-to-many expansion.                                                                |
| 9. PII protection                  | Hash passenger IDs and passport numbers; mask emergency-contact names; hash emergency-contact phones; remove raw passenger_id from the analytical booking fact. |

|                        |                                                                                                                                      |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| 10. Payment validation | Create/retain payment_valid and distinguish valid from invalid or missing payment records. Result: 922 valid and 78 invalid/missing. |
| 11. Gold modeling      | Create fact and dimension datasets for analytical reporting.                                                                         |
| 12. Validation/export  | Validate counts and KPIs, then export Gold CSV datasets for Power BI.                                                                |

## 7. Data Cleaning Strategy

- Business-key integrity: flight\_id is treated as the operational flight business key.
- Duplicate handling: exact duplicates are removed; duplicate flight IDs are resolved using the implemented earliest-departure rule.
- Missing/invalid values: fields are checked and controlled quality indicators are retained for reporting.
- Type standardization: timestamps, dates, durations and payment amounts are converted to analysis-ready types.
- Join integrity: passenger mapping is deduplicated before joining to bookings.
- PII minimization: sensitive passenger identifiers are protected before analytical consumption.

## 8. Duration Calculation and Overnight Flight Handling

- Duration is measured in minutes using departure and arrival timestamps. The calculation accounts for the calendar-day boundary so that an arrival after midnight is not interpreted as an earlier time on the same day.
- A flight is marked is\_overnight when its arrival date is later than its departure date. The final pipeline identified 122 overnight flights. One duration correction was recorded, and zero extreme-duration anomalies remained after validation.
- The source does not contain a dedicated delay-minutes field. Therefore, no synthetic delay value is created. The anomaly/quality dashboard instead reports duration corrections, overnight flights, duplicate removal and payment/data-quality indicators.

## 9. Final Data Quality Results

| Metric                        | Result |
|-------------------------------|--------|
| Raw flights                   | 1,020  |
| Clean flights                 | 1,004  |
| Flight duplicate rows removed | 16     |

|                            |                |
|----------------------------|----------------|
| Routes                     | 30             |
| Airlines                   | 6              |
| Average flight duration    | 164.54 minutes |
| Overnight flights          | 122            |
| Duration corrections       | 1              |
| Extreme duration anomalies | 0              |
| Bookings                   | 1,000          |
| Valid payments             | 922            |
| Invalid/missing payments   | 78             |
| Total revenue              | 7,385,142.98   |

## 10. Assumptions

- flight\_id is the primary business key for flight records.
- For duplicate flight IDs, the earliest departure record is retained.
- Overnight means arrival date is later than departure date.
- Duration is measured in minutes and respects midnight/day boundaries.
- Invalid or missing payment records remain available for data-quality analysis but are excluded from Total Revenue.
- No delay metric is fabricated because the source has no dedicated delay-minutes field.
- Gold datasets are the reporting layer consumed by Power BI.

## 11. Privacy and Access Control

- Passenger IDs and passport numbers are hashed; emergency-contact names are masked; emergency-contact phones are hashed; raw passenger\_id is removed from the analytical booking fact.
- Fine-grained individual/group RBAC was not implemented in Databricks Free Edition. The proposed production model is least-privilege access: Data Engineers can read/write Raw, Silver and Gold; Business Analysts and Power BI consumers receive Gold read-only access; Raw/Silver remain restricted.

## 12. Power BI Outputs

- Executive Operations Dashboard – operational KPIs, airline performance, route traffic and booking status.
- Duration Analysis – average duration, overnight flights, duration corrections and duration comparisons.
- Airline & Revenue Trends – flights, bookings, revenue/payment method and payment validation.

- Delay & Anomaly Insights – duration corrections, overnight flights, duplicate removal and data-quality indicators.

### **13. Reproducibility and Repository Contents**

The GitHub repository contains the Databricks notebook, Power BI .pbix report, Gold analytical CSV datasets, KPI/quality summaries and dashboard screenshots. The original source Excel workbook is intentionally not included in the public repository.